ROUND 2 OFFICIAL BUILD DOSSIER

Enterprise Brand Governance with Deterministic Authority Resolution

Multi-format compliance verification across Text, Image, and PDF collateral using Pure Java institutional hierarchies and Neon pgvector semantic RAG.

LIVE APPLICATION
https://claimledger.vercel.app

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SOURCE REPOSITORY
github.com/Ravikiranswamy/ClaimLedger

The Enterprise Brand Drift Challenge

Why generic LLMs and manual legal reviews fail modern enterprise scale.

THE VELOCITY TRAP

Explosion of Assets

Modern enterprises produce thousands of social ads, pitch decks, partner proposals, and banners weekly across dozens of decentralized marketing and sales teams.

THE LIABILITY RISK

Outdated & Conflicting Claims

Sales quotes superseded pricing (e.g., ₹799 vs official ₹999), marketing publishes unapproved performance claims, and unapproved logos or off-brand colors leak to public channels.

THE LLM FAILURE

Why AI Wrappers Fail

Pure LLMs hallucinate, produce non-deterministic verdicts across runs, and have zero understanding of institutional authority tiers (Tier 1 Master vs Tier 2 Exception).

Deterministic Authority Resolution

Decoupling language understanding from compliance truth.

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// Institutional 6-Step Rule Execution (Pure Java) Step 1: Status Filter → Discard non-approved or draft documents. Step 2: Temporal Check → Check effective & expiry dates (detect outdated). Step 3: Scope Match → Match product line, department & region. Step 4: Tier Hierarchy → Tier 1 (Company Master) > Tier 2 (Department) > Tier 3. Step 5: Numeric Match → Precise arithmetic comparator (₹799 ≠ ₹999). Step 6: Escapement → Flag discrepancies to Reviewer with exact proof.
```

*We never ask an LLM "Is this claim compliant?". Compiled Java code deterministically computes compliance based on institutional authority hierarchy.

Dual-Stream Processing Architecture

- ✓ **Text & In-Image Extraction:** Google Gemini 3.8 Flash isolates discrete factual assertions and character offsets.
- ✓ **Vector Retrieval (Neon pgvector):** Sub-5ms cosine similarity lookup against 768-dim embeddings in Singapore.
- ✓ **Deterministic Rule Engine:** Pure Java execution ensures 100% repeatability with zero hallucinations.
- ✓ **Append-Only Audit Ledger:** Every claim, extraction, revision, and reviewer sign-off is logged chronologically.

Multi-Format Content & Visual Governance

Extending brand governance beyond plain text to Images and PDFs.

VISUAL ASSETS KNOWLEDGE BASE

Visual Brand Assets (ASSETS)

- ✓ Official corporate logos & mark variations.
- ✓ Approved color swatches (#0A1F36, #0F766E).
- ✓ Typography, isolation clear-space rules.
- ✓ Stored with org ID, version, and Tier 1 authority.

MULTIMODAL ANALYSIS

In-Image OCR & Logo Checks

- ✓ Extracts text and pricing inside images.
- ✓ Detects unapproved competitor marks (Google, Amazon).
- ✓ K-Means dominant color extraction.
- ✓ Accessibility & clear-space adherence.

STRICT GOVERNANCE LOOP

No-Bypass Image Revision

- ✓ Text changes cannot resolve visual logo violations.
- ✓ Creator must replace image file with approved asset.
- ✓ 1-click Nexora Emblem test helper in UI.
- ✓ High-resolution Lightbox for reviewer sign-off.

Full Production Deliverables Completed

All planned modules built, integrated, and deployed live with zero stubs.

Knowledge & Engine Core

- ✓ **Multi-Tier Knowledge Base:** Full CRUD for Tier 1, 2, and 3 documents with department scoping.
- ✓ **pgvector Chunk Inspector:** Real-time inspection of 768-dim embeddings in Neon PostgreSQL.
- ✓ **Deterministic Engine:** 6-step rule hierarchy in pure compiled Java code.
- ✓ **Numeric Comparator:** Robust currency and frequency normalization (₹, $, €, /mo).

Workflow & Governance UI

- ✓ **Multi-Format Submission:** Paste Text, Upload Image (PNG/JPG), and PDF parsing.
- ✓ **Separated Findings Tabs:** Factual text claims vs Visual Brand Governance findings.
- ✓ **Reviewer Console:** Lightbox image inspection, audit timeline, and formal sign-off.
- ✓ **Immutable Audit Trail:** Append-only chronological event ledger with CSV export.

Production Infrastructure Stack

Engineered for sub-100ms response times and enterprise data isolation.

FRONTEND

React 19 & Vite 8

Tailwind CSS v4, TanStack Query, single-page reactive highlights. Deployed on Vercel Global Edge Network.

BACKEND

Spring Boot 3.3.5

Java 17, Spring Security + stateless JWT, Apache PDFBox, ImageIO. Dockerized Web Service on Render Cloud.

DATABASE

Neon pgvector

Serverless PostgreSQL in AWS Singapore (sub-5ms co-located latency). 768-dim IVFFlat cosine similarity.

24/7 AVAILABILITY

Automated Cron

Automated 10-min keep-alive ping on /api/health ensures zero sleeping delays and instant demo logins.

Three Substantial Round 3 Commitments

Expanding ClaimLedger from a web app into an enterprise-wide governance fabric.

IMPROVEMENT 1

Advanced Visual Heatmaps & WCAG Accessibility

Interactive visual coordinate bounding boxes ([x, y, w, h]) overlaid on promotional canvas.

- ✓ Automated WCAG AA/AAA contrast ratio verification.
- ✓ Font-size disclaimer legality verification.
- ✓ Visual heatmap highlighting off-brand clutter.

IMPROVEMENT 2

Multi-Jurisdiction Branching Conflict Tree

Simultaneous evaluation of collateral across international regional jurisdictions and sales channels.

- ✓ Cross-currency normalization (₹ India vs $ US vs € EU).
- ✓ Channel-specific rules (Social vs Reseller vs Partner).
- ✓ Automated semantic auto-patching of regional copy.

IMPROVEMENT 3

Real-Time Pre-Flight Browser SDK & Extension

Shift-left governance from "post-creation review" to "in-workflow prevention".

- ✓ Chrome Extension for Google Docs, Figma & Canva.
- ✓ Webhook listeners for Contentful & WordPress CMS.
- ✓ Live inline pre-flight warnings before export.

Submission Ready for Qualification

CERTIFICATE PRINTING DETAILS

Your Role on the Project:

Full-Stack Architect & AI Systems Engineer

The Part You Built (In Your Own Words):

"Engineered the complete ClaimLedger platform end-to-end, including the React 19 governance UI, Spring Boot backend, Neon pgvector semantic retrieval, and the pure Java 6-step deterministic authority resolution engine."

SUBMISSION CHECKLIST

- ✓ **Functioning URL:** https://claimledger.vercel.app
- ✓ **Source Code Repo:** github.com/Ravikiranswamy/ClaimLedger
- ✓ **Swagger API Docs:** claimledger-backend.onrender.com/swagger-ui.html
- ✓ **Database:** Serverless Neon PostgreSQL pgvector (Singapore)
- ✓ **Round 2 Feature Complete:** Text, Image & PDF multi-format visual governance operational.